

Part A · Multiple Choice

2 points each

1. **a) Mature stage**

p. 106 — this is when the storm produces strong gusting winds, heavy precipitation, lightning, and possibly hail. Once downdrafts overtake the updrafts, the storm enters the dissipating stage.

2. **a) 20%**

p. 106 — the other 80% evaporates back into the atmosphere.

3. **b) Fujita Scale**

p. 107 — it assigns a value based on wind speed and damage. The Saffir-Simpson Scale is for hurricanes.

4. **c) 28°C (82°F)**

p. 108 — they form between about 10° and 25°N in summer and autumn. If wind shear is low, a tropical depression can build into a hurricane in two to three days.

5. **b) eyewall**

pp. 108–109 — the eye itself is the relatively calm center. The eyewall is the ring of cloud immediately surrounding it.

6. **d) Polar (E)**

p. 111 — the five main Köppen groups are Tropical (A), Dry (B), Temperate (C), Continental (D), and Polar (E), each divisible into subcategories.

Part B · True or False

1 point each

7. **TRUE**

pp. 95 and 110 — the World Meteorological Organization uses 30 years as the standard averaging period. Climate also takes in the weather extremes for a place, not just the averages.

8. **FALSE**

Cyclones rotate counterclockwise around a low-pressure center in the Northern Hemisphere, and clockwise in the Southern Hemisphere. (pp. 107–108)

Part C · Short Answer

3 points each

9. **Weather is the state of the atmosphere at a particular place and time; climate is the average of weather elements over a long period, generally 30 years.**

Rubric: 1 pt weather — the state of the atmosphere at some place and time, expressed through temperature, pressure, humidity, wind, precipitation, and cloudiness; it can change rapidly. 1 pt climate — the mean of those elements over a specified period (about 30 years), plus the extremes for that place; it changes only very slowly. 1 pt for the contrast in practice: meteorologists work with real-time data to forecast hours to weeks ahead and locally, while climate scientists look at averages and trends across decades, centuries, and millennia, using proxy data alongside weather records. (pp. 110–112)

10. A storm surge is the unusually high rise in sea level that occurs as a hurricane's low-pressure center comes onto land, worsened by high winds pushing seawater toward the shore.

Rubric: 1.5 pts for the mechanism — the storm's low-pressure center reaching land causes sea level to rise unusually high, and the hurricane's high winds blow seawater across the ocean onto the shoreline, making it worse. 1.5 pts for the consequence — flooding is devastating on low-lying coasts such as the Atlantic and Gulf Coasts, since there is no elevation to stop the water. Credit the Hurricane Michael example (2018, peak winds 160 mph, storm surge up to 14 feet). (p. 109)